Machine learning
in real companies

Predict, group and combine: from idealista and Spotify to the Uber simulator

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04. Putting it together: the Uber simulator

05. Conclusions and next steps

Three real cases, from simple to integrated. The demo figures are synthetic.

Objectives

What we will learn

Tell **predicting a value** (regression) apart from **discovering groups** (clustering) and what each task is for.

See how real companies do it: idealista values properties and Spotify segments users.

See how **several techniques combine** in one system: Uber dynamic pricing.

What you take away

A way to read any use case: **data → model → decision**.

A numeric example of each technique, to see how it works inside.

The limits of each approach: what data it needs, how it is validated and what can go wrong.

01

Prediction: idealista

First task: estimate a number from the characteristics of an observation

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Regression analysis

idealista/data (Spain)

Commercial automated valuation service

Concept

Estimate a continuous number from explanatory features.

Real use case

idealista/data combines comparables and hedonic ML to estimate property values from location and attributes.

Advantage

Produces consistent estimates across large property portfolios.

Limitation

Missing attributes and market changes reduce accuracy. Estimates differ from completed sale prices.

The company discloses the valuation approach, but not the exact estimator.

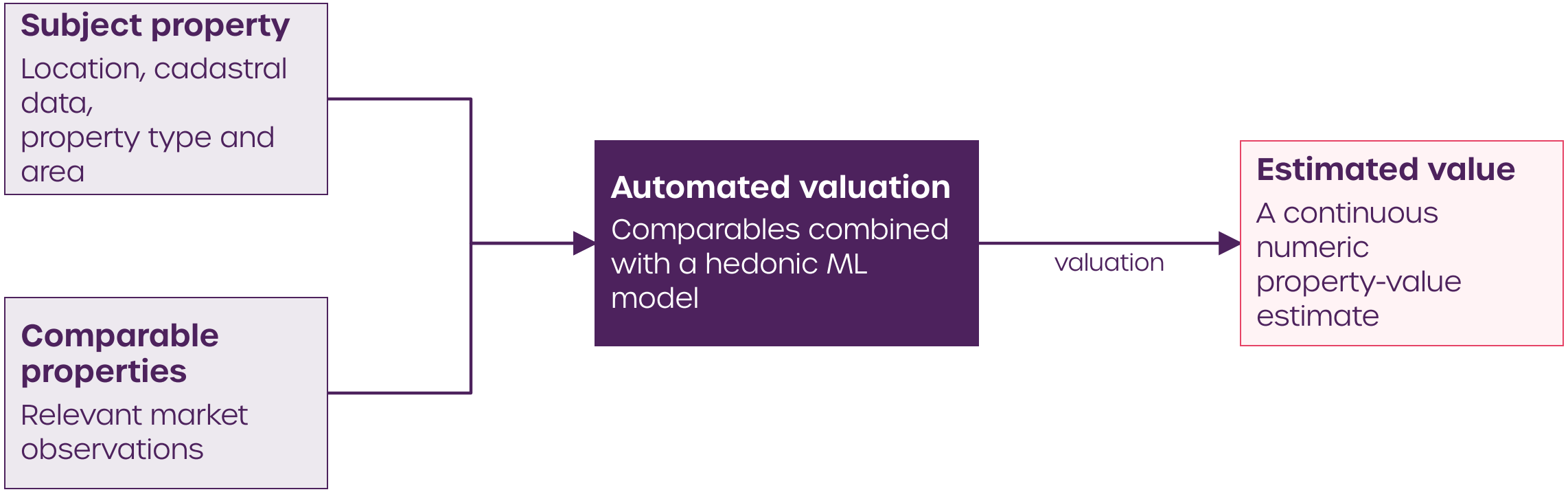
Regression estimates a number from the characteristics of an observation.

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data



Source: idealista/data and idealista labs

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idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

The subject property becomes a feature vector

Ana's apartment

Zone B · fictional market

Floor area	80 m²	Bedrooms	2
Renovated	Yes	Lift	Yes

↓

$$x = [80, 2, 1, 0, 1, 1]$$

area, beds, zone B, zone C, renovated, lift

Zone A is the reference. Zones B and C use separate 0/1 indicators.

Computed demo · synthetic homes and prices

1 / 5

Technical details

Ana's home to value

Ana wants to estimate her apartment's price. Change its attributes to follow the same home through every step.

Floor area — 80 m²

Bedrooms — 2

Fictional zone — Zone B

Renovated — ☑

Lift — ☑

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

Comparable evidence from the training set

Zone B only · 80 m² · 2 bedrooms · six closest matches

Property	Area	Beds	Renovated	Lift	Reference price
P425	85 m²	2	Yes	Yes	€302,180
P276	91 m²	2	Yes	Yes	€339,450
P201	80 m²	3	Yes	Yes	€311,444
P393	80 m²	1	Yes	Yes	€283,257
P275	85 m²	3	Yes	Yes	€311,032
P216	71 m²	2	No	Yes	€278,683

Mean of these six prices **€304,341**

All properties and prices are synthetic. Matching rules are for teaching.

Computed demo · synthetic homes and prices

2 / 5

Technical details

Six similar reference properties

Find six homes like Ana's: the same zone, similar size and bedrooms. Their reference prices provide a first comparison.

What makes a match?

Same zone, similar area and bedrooms, plus renovation and lift.

Only training properties enter this comparison.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

Learning first, prediction second

TRAINING · computed once

360 known examples
Features X + price y

→ **Linear regression**
Learn weights β

→ **OLS fit**
$\min \Sigma(y - \hat{y})^2$

PREDICTION · updates with your inputs

This home's features
80 m² · Zone B

→ **Fixed fitted model**
$\hat{y} = \beta_0 + \Sigma \beta_j x_j$

→ **Model price**
€305,790

Computed demo · synthetic homes and prices

3 / 5

Technical details

Regression learns a numeric relationship

Learn from known homes, then predict Ana's price from her apartment's attributes.

What does it learn?

For example, +10 m² adds €19,802 to this model's prediction, holding other inputs fixed.

This is a fitted association in synthetic data.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

A transparent blend for this demo

Similar-home mean

€304,341

50% of the demo estimate

+

Regression estimate

€305,790

50% of the demo estimate

Illustrative estimated price

€305,000

Rounded to the nearest €1,000

The 50/50 blend is our teaching choice. idealista's production weighting is not public.

Computed demo · synthetic homes and prices

4 / 5

Technical details

One continuous price estimate

For Ana's home, combine the similar-home mean with the regression prediction. This demo uses an illustrative 50/50 average.

Held-out mean absolute error

€10,577 across 120 synthetic homes.

A demo evaluation, not idealista's accuracy or an individual prediction interval.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

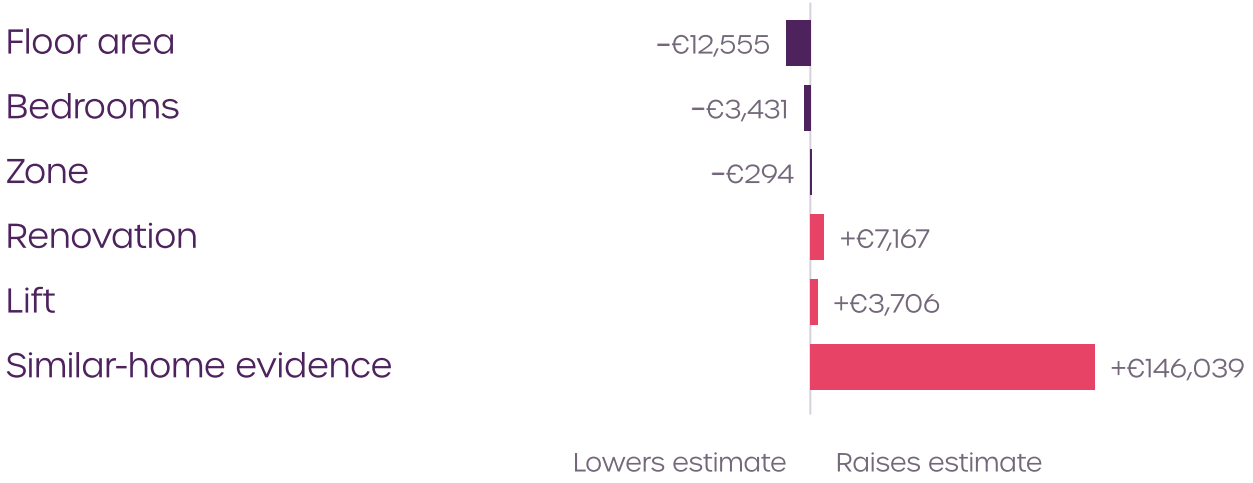
idealista: estimating a property value

Property attributes and comparable properties feed automated valuation

idealista/data

1 Example | 2 Similar homes | 3 Regression | 4 Estimate | 5 Explainability

An arithmetic breakdown of this estimate



Baseline €316,604 + adjustments = €305,065

Exact demo arithmetic, before rounding. The baseline is the mean training-model prediction.

Computed demo · synthetic homes and prices

5 / 5

Technical details

What problem does this solve?

For Ana, explain how size, renovation and lift affect this estimate.

This apartment

80 m² versus a training average of 93 m²: floor area contributes −€12,555.

Renovation and lift

Renovated: +€7,167. Has a lift: +€3,706.

Bars explain this calculation. They do not prove causes or guarantee a sale price.

← / → steps · teaching baseline · no idealista data or proprietary model

Source: idealista/data and idealista labs

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02

Classification and clustering: Spotify

Second task: discover groups when there are no prior labels

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Clustering

Spotify (Sweden)

Documented user-research method

Concept

Discover groups with similar behavior without predefined group labels.

Advantage

Finds unexpected segments and supports targeted user research.

Real use case

Spotify combines UMAP with recursive HDBSCAN to discover finer segments of users.

Limitation

Feature choices shape the groups. Stable clusters still need human validation.

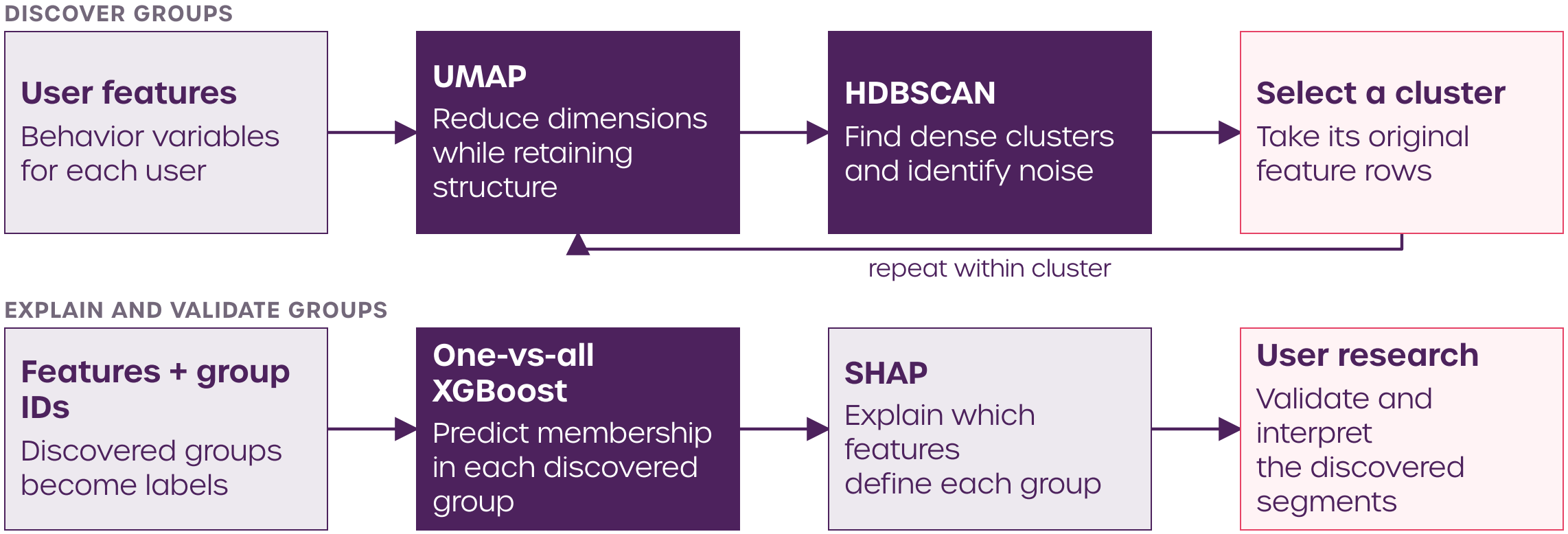
Spotify uses XGBoost and SHAP to explain the resulting segments.

Clustering discovers groups that researchers can then investigate.

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster



Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

The people behind the use case

People	Ana and Luis play AC/DC and Metallica. Marta plays Taylor Swift.
Group	Artist-play counts could group Ana and Luis together, apart from Marta.
Why	An explanation could identify rock and metal listening as the group's main signal.
Possible use	Suggest Iron Maiden to Ana if Luis listens to them.

Illustrative artist example. The computed demo uses behaviour, not artist counts.

1 / 9

Technical details

A concrete listening example

Ana and Luis share artists. Can we discover their group and explain what makes it different?

Follow the same people

Each next step explains what the algorithm would mean for Ana, Luis and Marta.

Illustrative story. The calculated charts use synthetic behaviour features, without artist counts or named users.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Original features

$X \in \mathbb{R}^{684 \times 6}$

Unlabelled observations

Listener	Minutes	Sessions	Skips	Discovery	Playlists	Night
L001	222	11.4	10%	17%	52%	72%
L002	124	6.6	46%	64%	41%	29%
L003	32	1.8	28%	27%	75%	8%
L004	127	6.6	42%	66%	32%	36%

Numeric features, no prescribed segment labels

684 rows × 6 features

StandardScaler $z_{ij} = (x_{ij} - \mu_j) / \sigma_j$ Fit once on the full demo population

Computed demo · synthetic listeners

2 / 9

Technical details

An original feature vector for every listener

Ana's listening becomes a numeric row. Here we compute six behaviour features. Artist counts would be extra inputs.

Listener L001

Minutes	222	Sessions	11.4
Skips	10%	Discovery	17%
Playlists	52%	Night	72%

← / → steps · colours identify local groups · no Spotify data or internal parameters

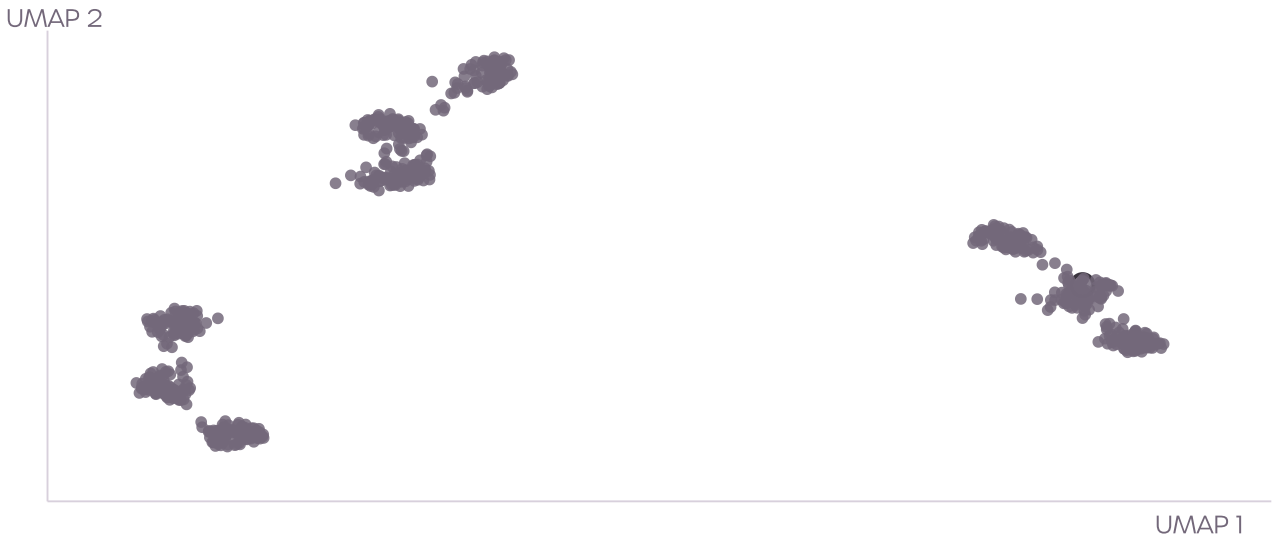
Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



Display transition only. Point movement does not reproduce the optimizer's iterations.

3 / 9

Technical details

UMAP makes neighbourhoods visible

With suitable artist features, Ana and Luis could appear nearby because they share artists. UMAP displays similarity, not a genre label.

684 × 6 → 684 × 2

10-neighbour trustworthiness: 0.975

Local-neighbourhood diagnostic, not segment validity.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Spotify: discovering and explaining groups

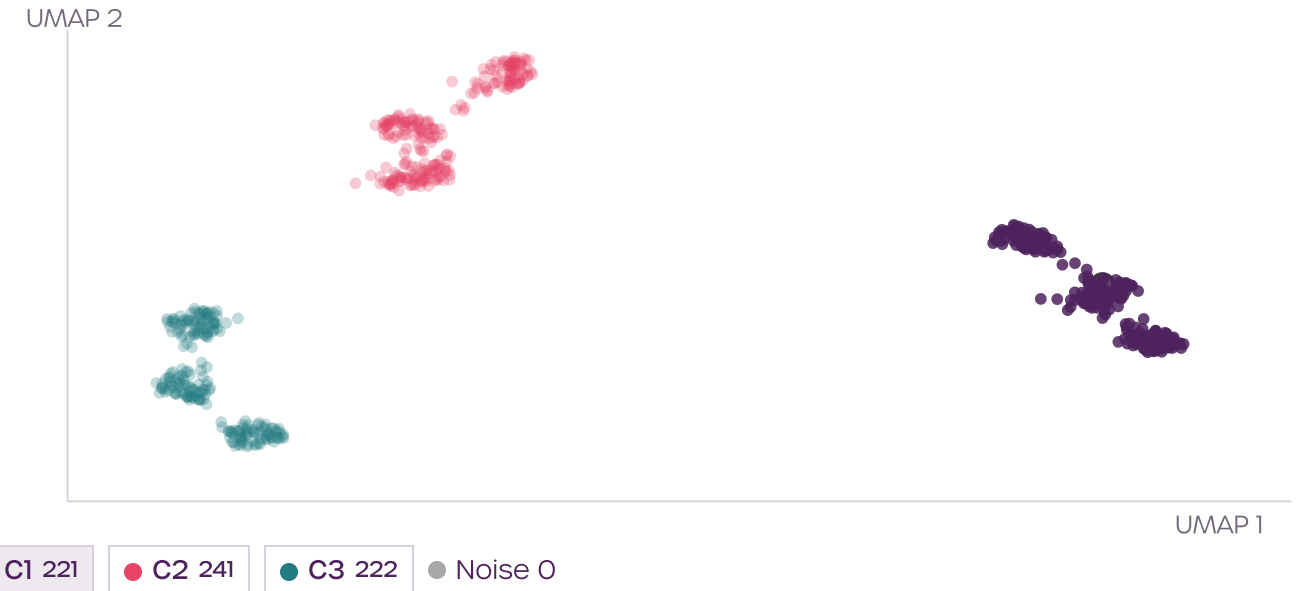
The recursion uses original features within a selected cluster

Spotify

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

Computed UMAP coordinates

3 clusters · 0 noise · 684 listeners



C1 221 | C2 241 | C3 222 | Noise 0

Computed demo · synthetic listeners

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Technical details

HDBSCAN discovers density-based groups

A dense group could contain Ana and Luis, while Marta falls elsewhere. The colours show discovered groups. Our chart uses synthetic behaviour data.

Inspect group: C1 · 221 listeners

32.3% of this node

Noise: 0 / 684

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

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Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

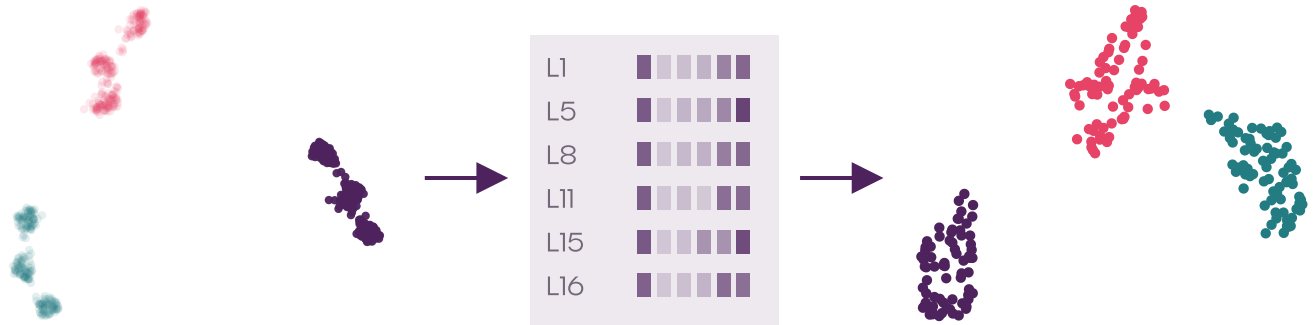
1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | 6 Hierarchy | 7 XGBoost | 8 SHAP | 9 Explainability

A new fit, not a magnified plot

1. Select C1

2. Original X rows

3. Refit both models



221 × 6 features

3 groups · 0 noise

New coordinates come from a new UMAP fit on the selected original rows.

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Technical details

Recursion starts from the original feature rows

Analyse Ana's group again using its original rows. This could reveal finer habits within her broader group.

Parent cluster: C1 · 221 rows

Inspect C1 child fit

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

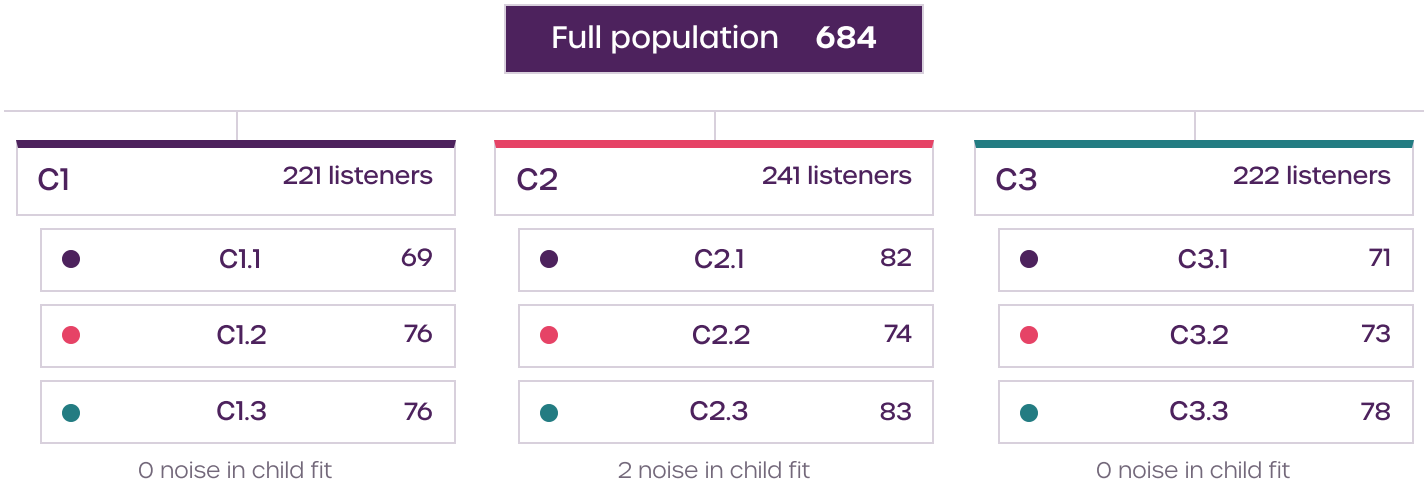
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Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

1 Example | 2 Features | 3 UMAP | 4 HDBSCAN | 5 Re-embed | **6 Hierarchy** | 7 XGBoost | 8 SHAP | 9 Explainability

Computed cluster hierarchy



Noise stays unassigned. Child counts can sum to less than their parent.

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Technical details

The hierarchy records each discovery step

Ana's broad group could split into subgroups. The tree records which users enter each new fit. Click a branch to inspect its computed population.

4 computed fits

Each child has its own UMAP coordinates and HDBSCAN assignments.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

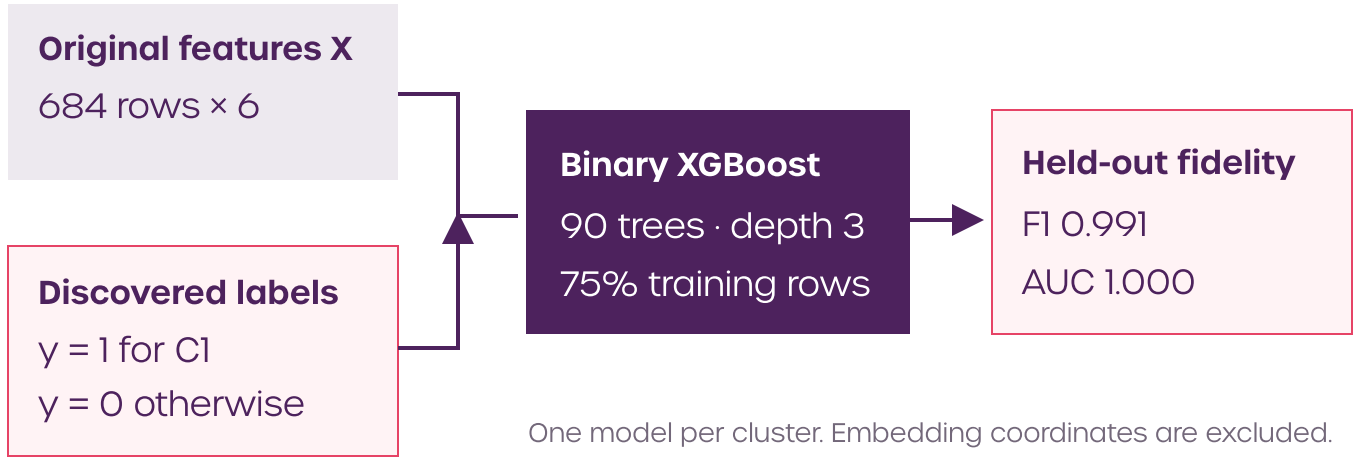
Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Spotify

One-versus-rest training



One model per cluster. Embedding coordinates are excluded.

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Technical details

A supervised model learns discovered membership

XGBoost learns to predict Ana's group versus everyone else, using the original features and discovered labels.

Membership model: C1 versus rest

513 training / 171 held-out

The split tests label reproduction, not new-user cluster discovery.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

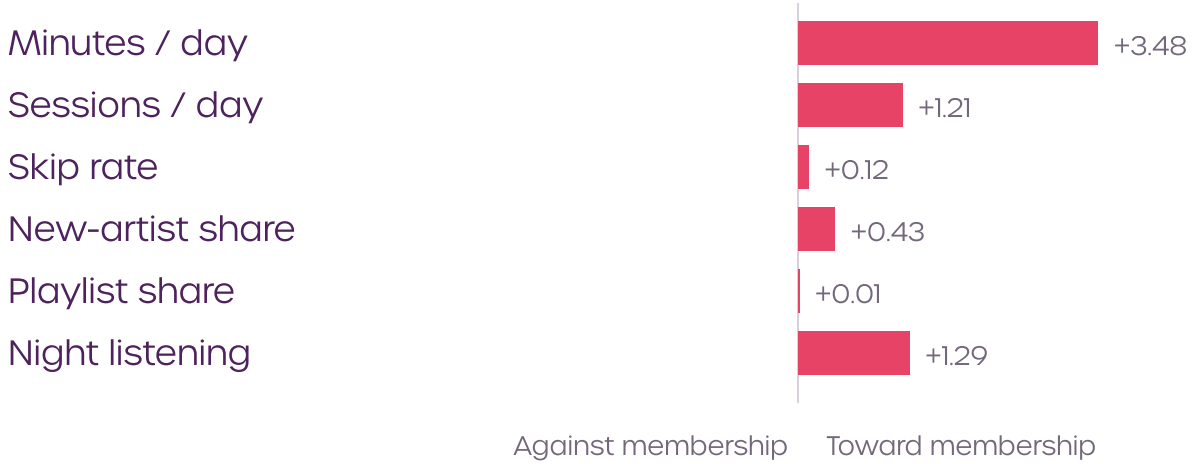
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Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Calculated TreeSHAP contributions

This listener | Mean |SHAP| | Signed contribution to membership log-odds



Base -0.80 + Σ SHAP 6.55 = log-odds 5.75

P(C1) = 99.7%

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Technical details

SHAP explains the membership predictor

SHAP explains a listener's group score. Indie listening could matter with artist inputs. These bars explain behaviour.

Explain membership in: C1

Listener: L001 · held-out

Held-out listener. Independent binary scores need not sum to 1.

Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

Spotify: discovering and explaining groups

The recursion uses original features within a selected cluster

Why interpretation is needed

People	Ana and Luis play AC/DC and Metallica. Marta plays Taylor Swift.
Group	Artist-play counts could group Ana and Luis together, apart from Marta.
Why	An explanation could identify rock and metal listening as the group's main signal.
Possible use	Suggest Iron Maiden to Ana if Luis listens to them.

Illustrative artist example. The computed demo uses behaviour, not artist counts.

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Technical details

What the results mean

Group C1 Listener L001

Across listeners

Minutes / day and Night listening have the largest average effects.

This listener

Minutes / day = 222 is the strongest push toward C1.

Model explanations do not prove causes or validate segments.

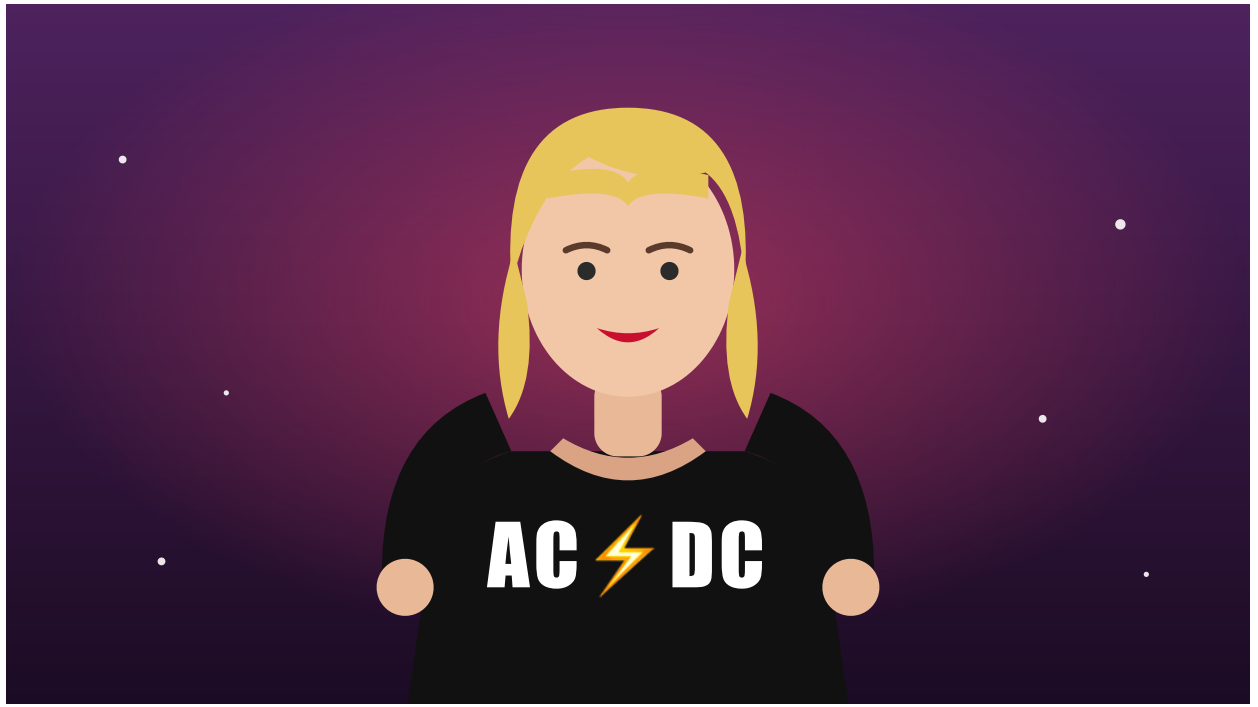
Computed demo · synthetic listeners

← / → steps · colours identify local groups · no Spotify data or internal parameters

Source: Spotify Engineering

When the group picks the wrong group

Marta listened to Taylor Swift, but if one day she puts on an AC/DC T-shirt... is she still in her group?



03

Putting it together: the Uber simulator

A real system does not use one technique, but several chained together

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From two cases to a complete system

What we have seen

Regression (idealista): estimate a continuous value from attributes.

Clustering (Spotify): group similar observations without labels.

What Uber adds

Time series: forecast the demand of the next minutes.

A control loop: correct the price every few minutes until supply and demand balance.

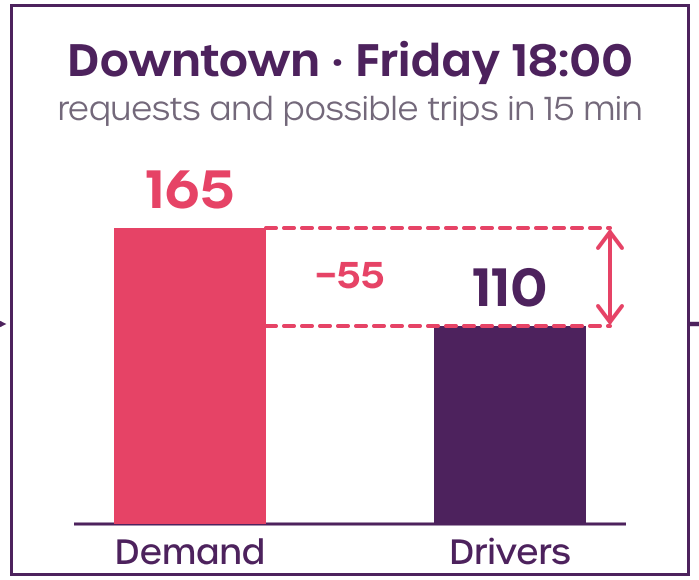
Uber: the problem

In every zone and every few minutes, demand and drivers almost never match

WHAT MOVES DEMAND

- Time and day
- Weather
- Events
- Traffic

THE ZONE RIGHT NOW



IF NOBODY ACTS

Demand above supply

Long waits, more cancellations and less satisfied users

WHAT HAPPENS IN DOWNTOWN

Supply above demand

Fewer trips per driver, less income and log-offs

IN ANOTHER ZONE OR AT ANOTHER TIME

Challenge: balance supply and demand within minutes, zone by zone

The challenge is not just to predict: it is to balance supply and demand before the gap hurts everyone.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Dynamic pricing and demand

Uber (global presence)

Explanatory case study; Uber's real models are proprietary

Concept

Predict demand per zone and adjust the price in real time.

Advantage

Shorter waits and a more stable driver supply.

Real use case

Time series, clustering and regression to set the dynamic price.

Limitation

Sharp surges feel unfair and require sensitive data.

The article publishes no real figures or architecture: the numbers in this deck are synthetic.

Price is a signal that balances the system, not just a charge.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

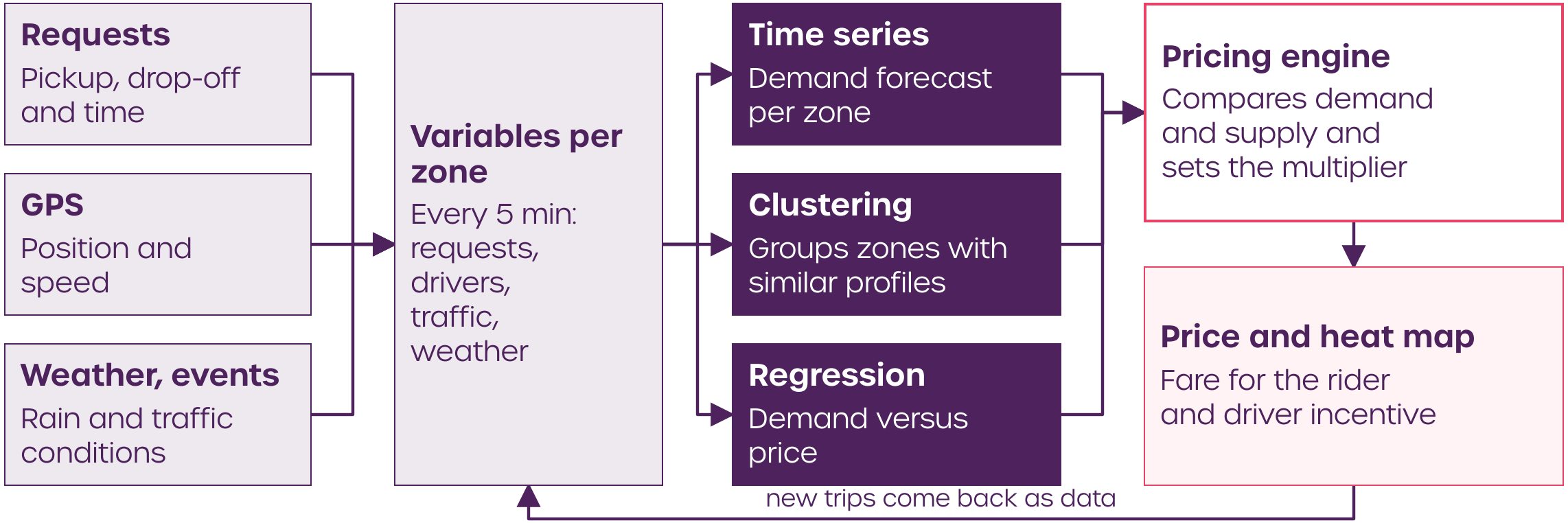
The starting data

Source	What it contains	What it is for
Requests and trips	Pickup, drop-off, time, duration, route, cancellations and payments	Measure real demand and its history
Phone GPS	Position and speed of each driver, every few seconds	Know supply and traffic in real time
External context	Weather, road closures and local events	Anticipate peaks that history does not explain
App usage	Ratings, cancellations and comments	Understand behaviour and satisfaction

Volume, variety and velocity: the three "V"s of Big Data. Source: Shuaib Ullattil, Medium (5 Feb 2026).

Uber: from data to price

From raw data to the price the rider sees, with a feedback loop



Didactic reconstruction based on the article's description: Uber's real architecture is not public.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Algorithms

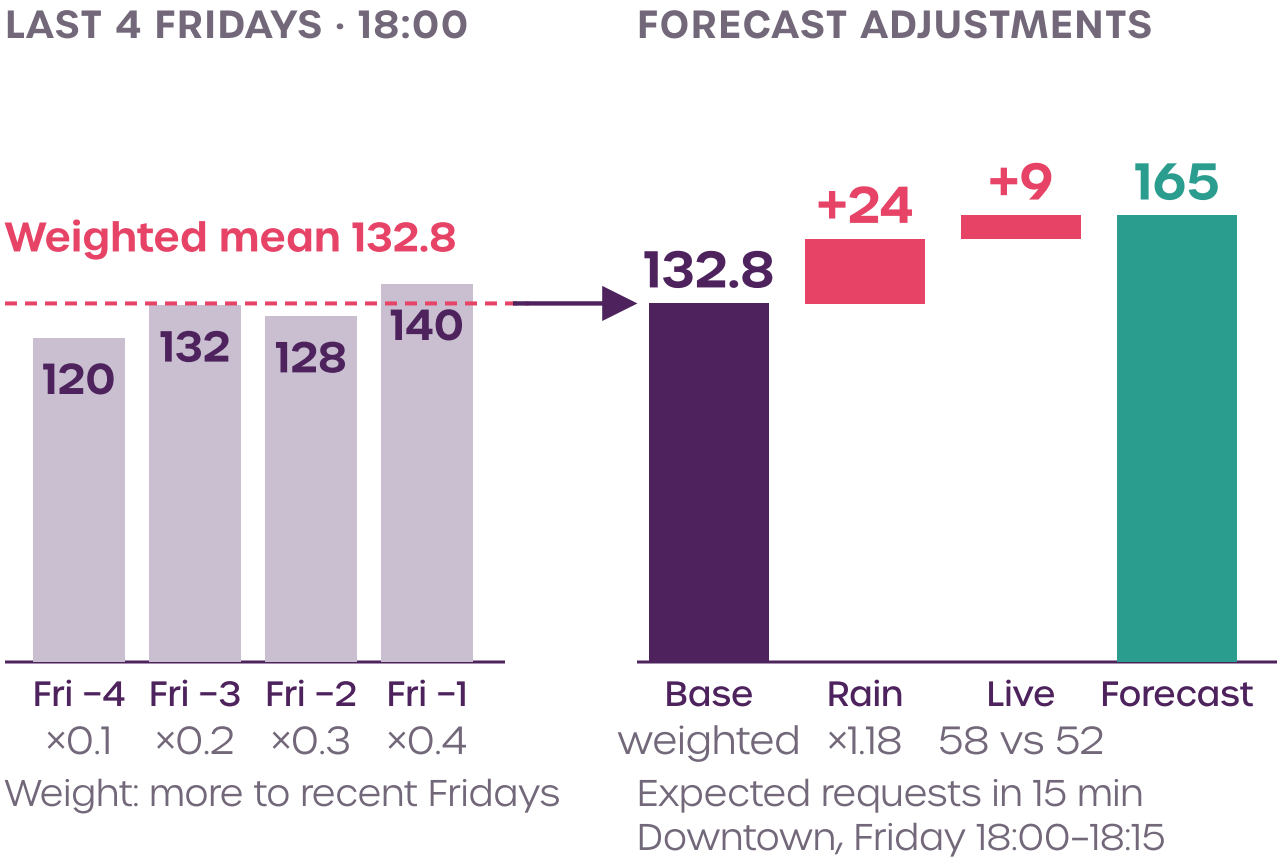
Four pieces, step by step

Each one with a synthetic numeric example: Downtown, Friday 18:00, with rain.

Time series: how many trips will there be?

Algorithm 1/4 Forecast of Downtown demand between 18:00 and 18:15

1. **Look at the past.** Requests at that time over the last 4 Fridays: 120, 132, 128 and 140.
2. **Weight.** Recent Fridays count more: 0.1·120 + 0.2·132 + 0.3·128 + 0.4·140 = **132.8**.
3. **Adjust for context.** A model learned that rain adds ≈18% demand: 132.8 × 1.18 = **156.7**.
4. **Correct live.** In the first 5 min there are 58 requests against 52 expected (+11%). Half is applied: 156.7 × 1.055 = **165**.

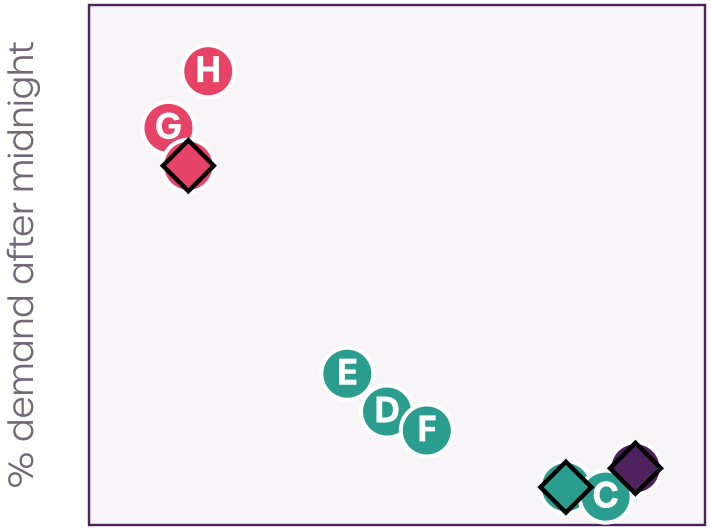


Synthetic data. Uber does not publish its models: a weighted seasonal average with adjustments is used here, the simplest form of a time series.

Clustering: what kind of zone is it?

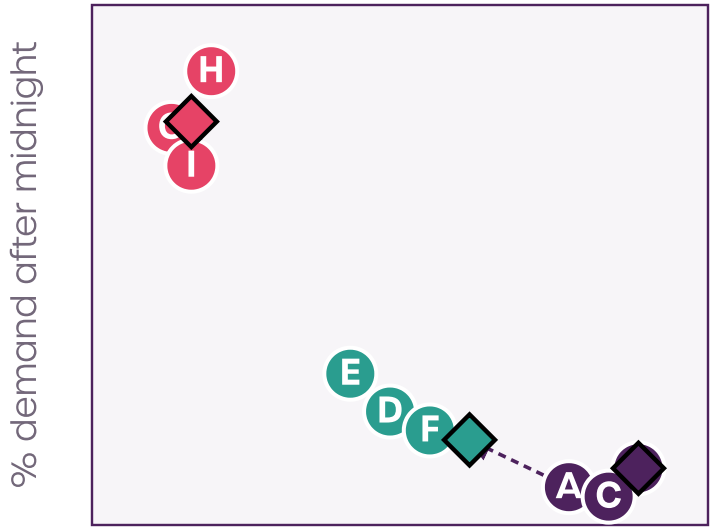
Algorithm 2/4 k-means groups zones by their demand profile (k = 3)

1 · Initial centres



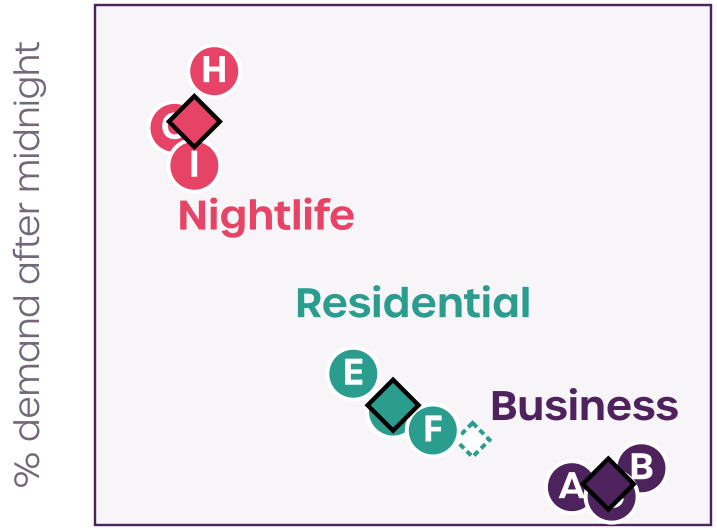
Two centres land in the same business zone. Each zone joins its nearest centre.

2 · Move and reassign



Each centre moves to the mean of its zones. A and C change group.

3 · Repeat until it converges



No zone changes: 3 groups remain. Centre (C) is a business zone.

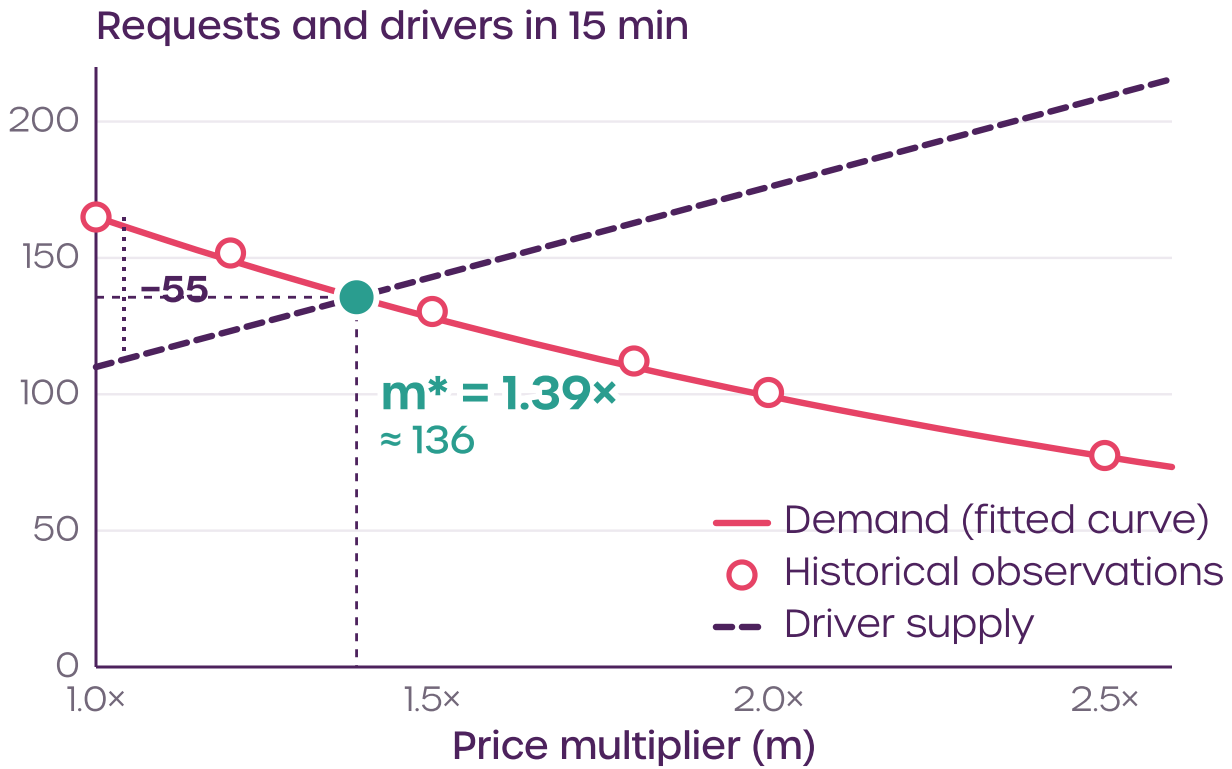
Synthetic data for nine zones (A–I). Each zone is described by two numbers; the groups emerge from the data, with no prior labels. Zones in the same group share a pattern: they are forecast together and it is anticipated where to send drivers.

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Regression: demand versus price

Algorithm 3/4 Sensitivity of demand to price and equilibrium price

1. **Gather (price, demand) pairs.** Six historical levels: at 1.0× demand is 100%; at 1.5×, 79%; at 2.5×, 47%.
2. **Choose the shape.** Demand falls exponentially: `demand(m) = base · e^(−β·(m−1))`.
3. **Fit by least squares** on ln(demand): **β = 0.51**, $R^2 = 0.999$. Each +1.0× removes ≈40% of demand.
4. **Find the equilibrium.** Demand(m) = supply(m): $165 \cdot e^{(-0.51(m-1))} = 110 \cdot (1 + 0.6(m-1)) \rightarrow$ **m* = 1.39×**, about 136 trips.



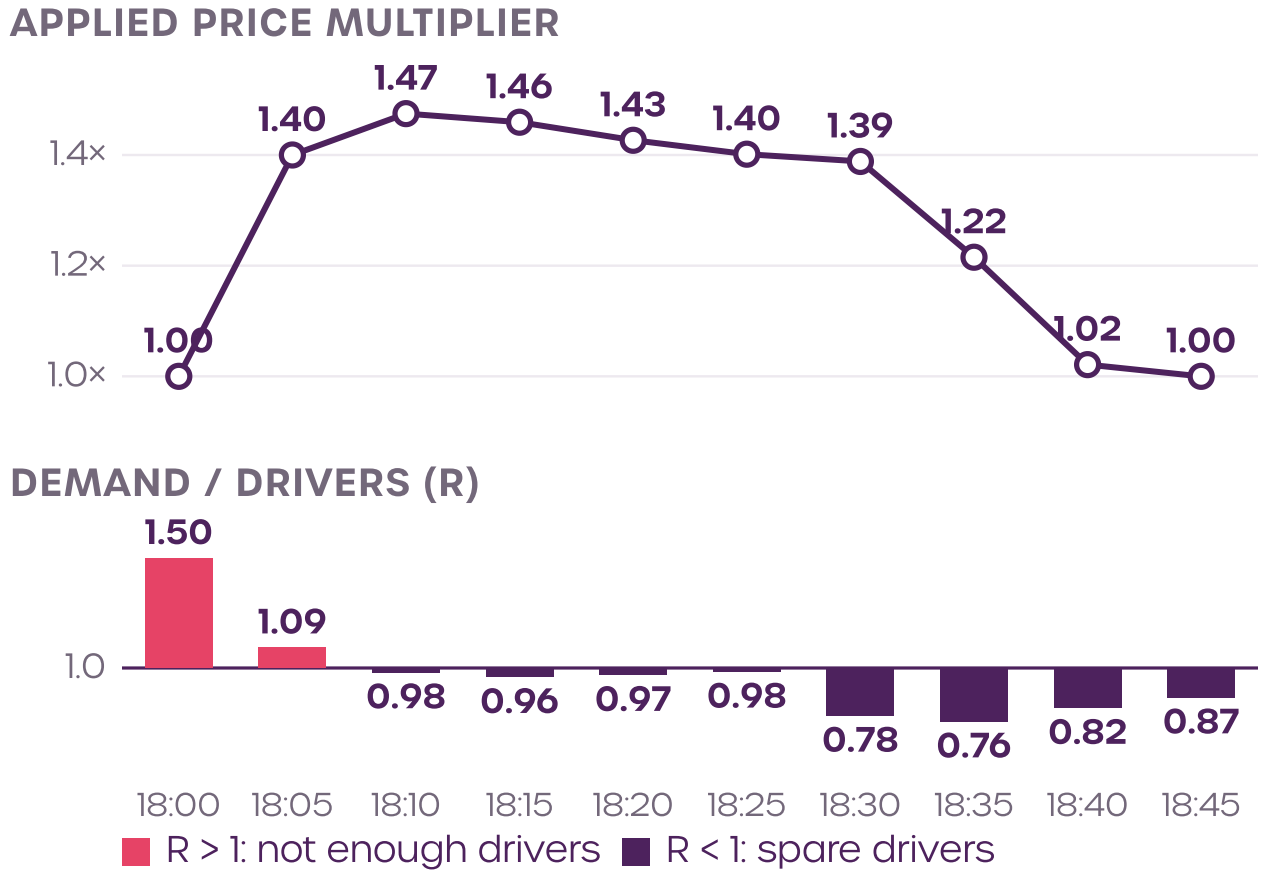
demand(m) = 165 · e^(−0.51·(m−1))
supply(m) = 110 · (1 + 0.6·(m−1))

Synthetic data. The supply response (+60% drivers for every +1.0×) is a didactic hypothesis; in practice it is also estimated from data.

Price loop: adjust every 5 minutes

Algorithm 4/4 Feedback control: the price is corrected every 5 minutes until it balances

1. **Measure.** Every 5 min: R = demand / drivers. At 18:00, 165 / 110 = **1.50**.
2. **Correct the price.** $m \leftarrow m + 0.8 \cdot (R - 1)$, between 1.0× and 3.0×: 1.00 + 0.8·0.50 = **1.40×** from 18:05.
3. **Let it react.** Some riders wait and drivers move closer. At 18:05, R = 1.09 → **1.47×**.
4. **Repeat and relax.** It settles at ≈1.4×. When demand falls (R < 1) the price drops to 1.00× at 18:45.



Synthetic simulation with the demand and supply of the example. When demand drops at 18:30, the multiplier returns to 1.00× on its own.

A Friday at 18:00, end to end

Downtown, Friday 18:00, with rain: the four algorithms chained with the example figures



Synthetic figures consistent with the previous slides: not Uber data.

Source: Shuaib Ullattil, Medium (5 Feb 2026)

Challenges and ethics

Privacy

Real-time locations of riders and drivers: they require consent, security and data governance.

Perception of fairness

Raising prices in emergencies or bad weather feels abusive, even if it balances the system.

Technical scale

Streams from millions of users with low latency; models are monitored and retrained.

Ideas to mitigate

A cap on the multiplier, a price shown before confirming and clear notices to the user.

Conclusions and next steps

Conclusions

Regression estimates a number, but does not guarantee a sale price.

Clustering discovers groups without labels, and the groups need validation.

Real systems combine techniques: Uber chains time series, clustering, regression and a control loop.

Next steps

Jupyter notebooks to simulate each case step by step.

A web app that simulates the behaviour of Uber dynamic pricing.

Try the ideas with your own data and measure the impact on a specific decision.

Sources

idealista/data Regression analysis

idealista/data · 17 Feb 2021

Spotify Clustering

Spotify Engineering · 5 Dec 2023

Uber Dynamic pricing and demand

Shuaib Ullattil, Medium · 5 Feb 2026

Explanatory or company sources. The demo data and the Uber example are synthetic. Consulted from 19 October to 10 November 2026.

Thank you

Which decision in your company could rely on one of these three approaches?

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